

# Engineering Curves: Construction of Cycloid

Rolling Circle Method with Tangent & Normal

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# Problem Statement

## Question

Draw a **cycloid** generated by a circle of diameter  $D = 50$  mm rolling along a straight base line for **one complete revolution** without slipping.

Construct a **tangent** and a **normal** to the cycloid at any point  $P$  located on the curve.

## Step 1: Draw Generating Circle & Directing Line

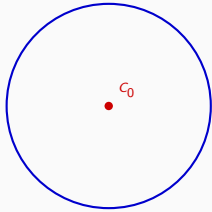
$C_0$  → Initial Center Point

$AA'$  → Directing Line

**Axis** → Center Line



## Step 1: Draw Generating Circle & Directing Line

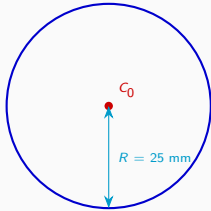


$C_0 \rightarrow$  Initial Center Point

$AA' \rightarrow$  Directing Line

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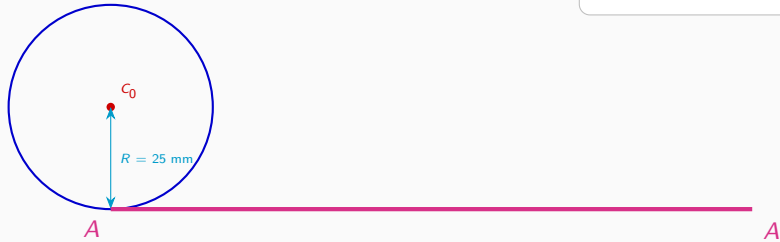


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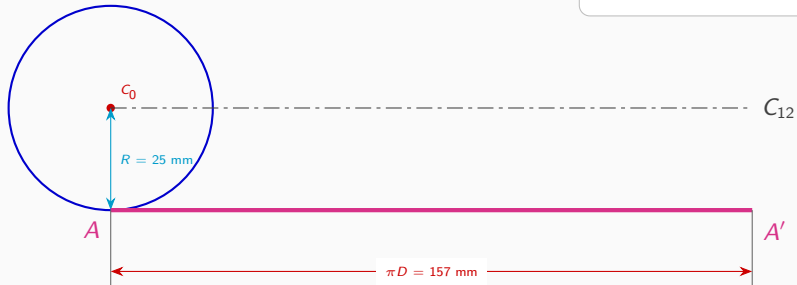


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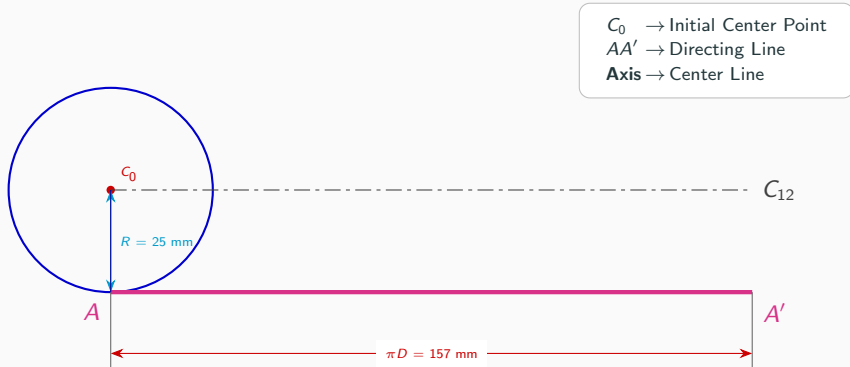


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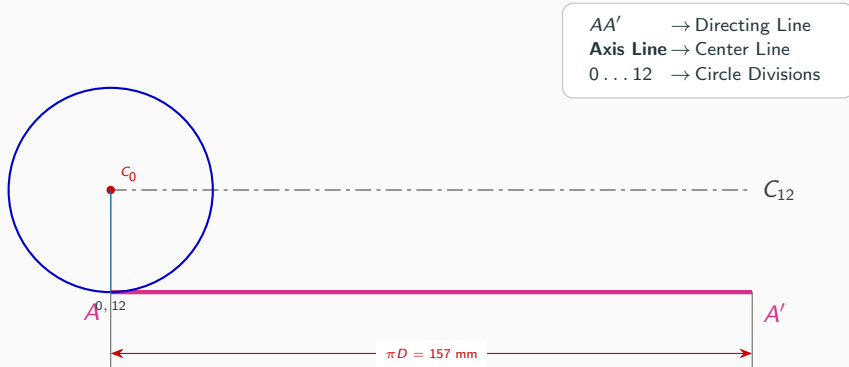
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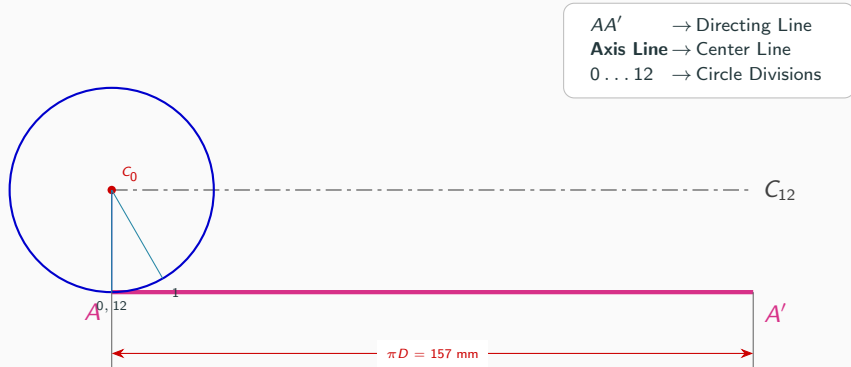
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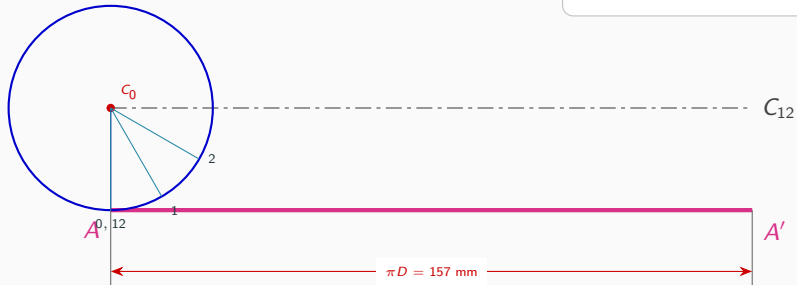
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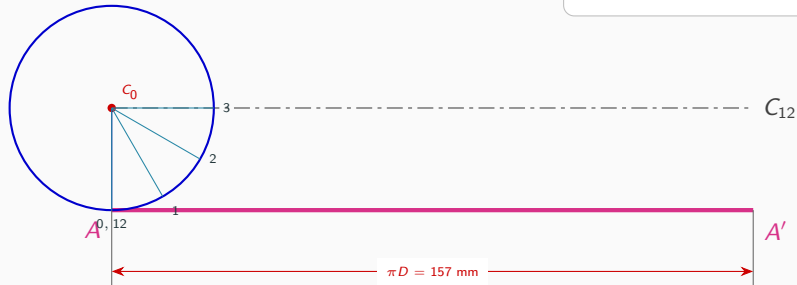
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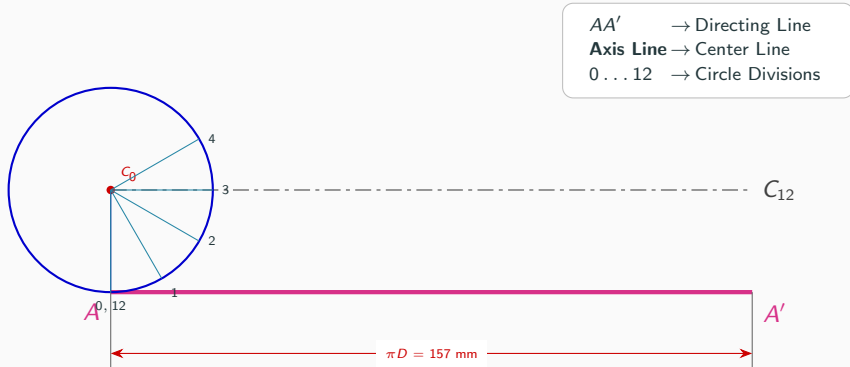
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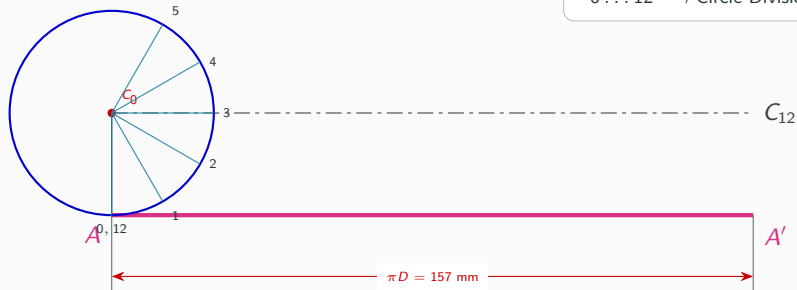
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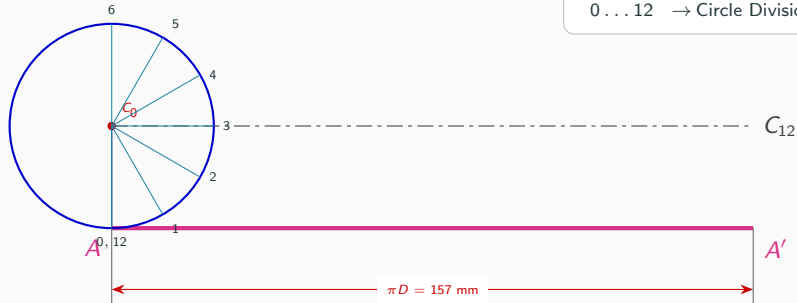
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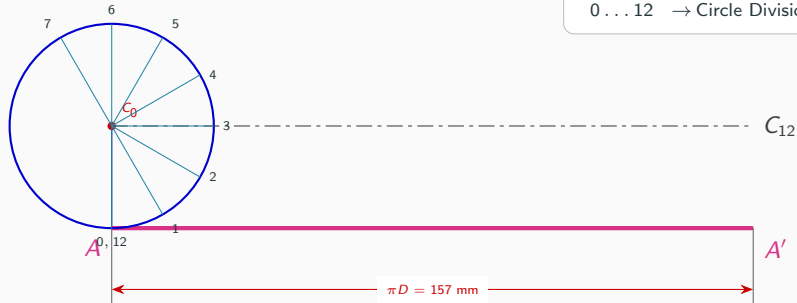
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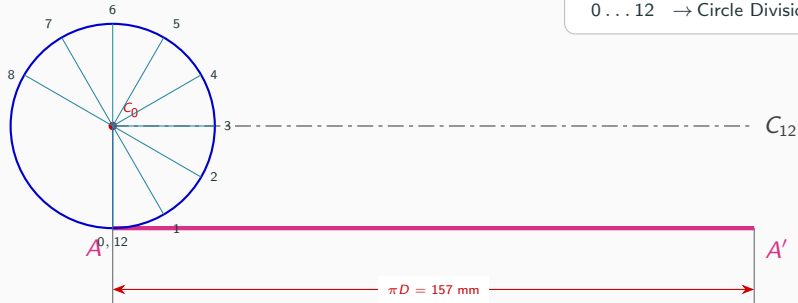
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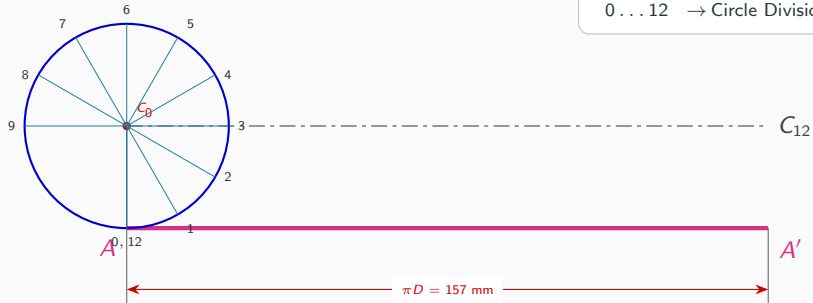
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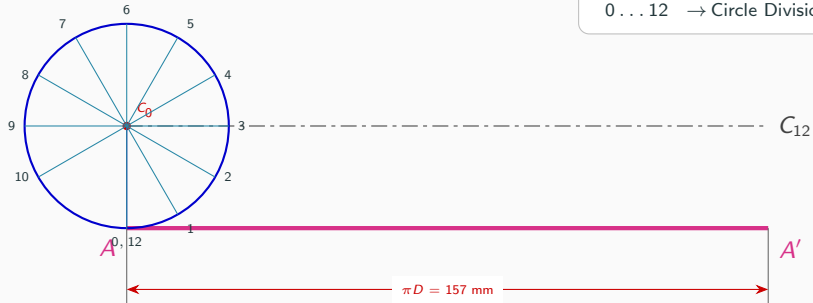
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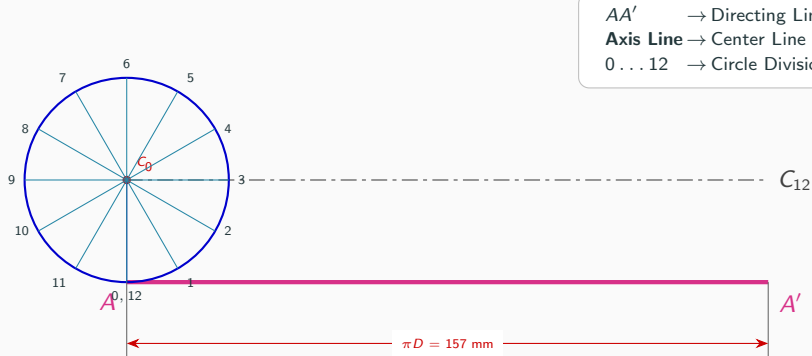
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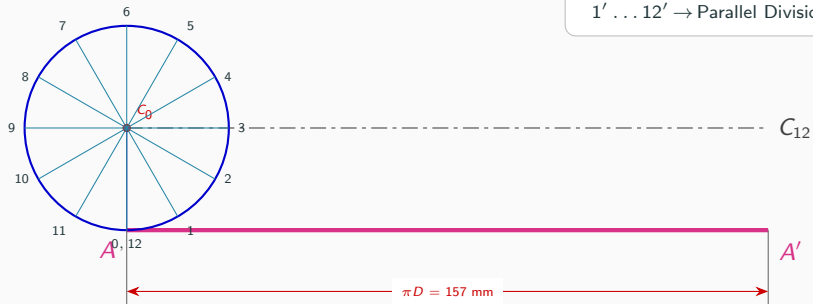
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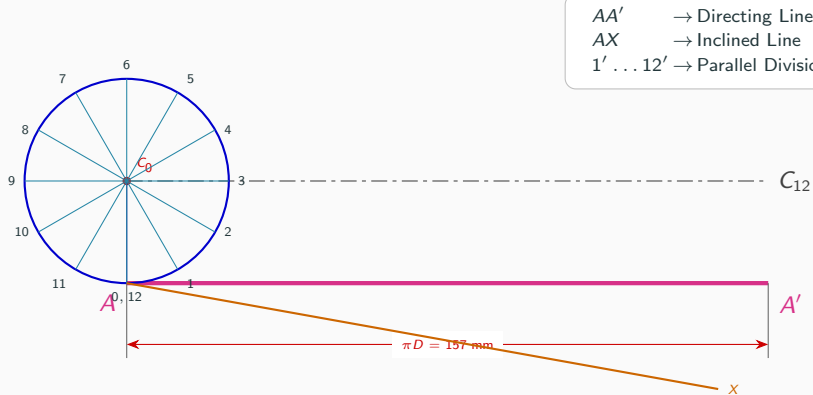
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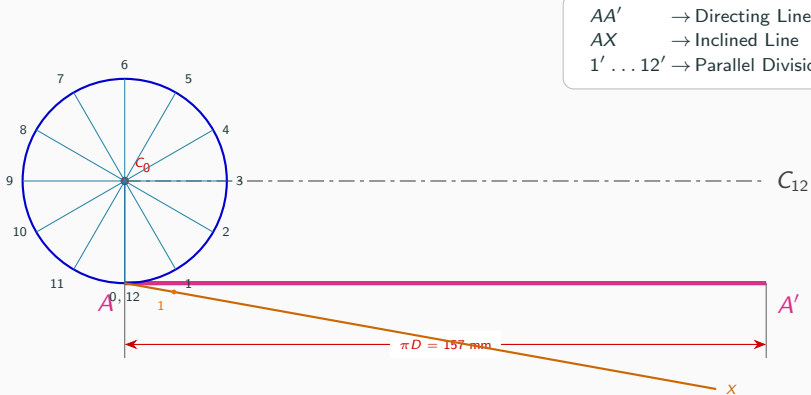
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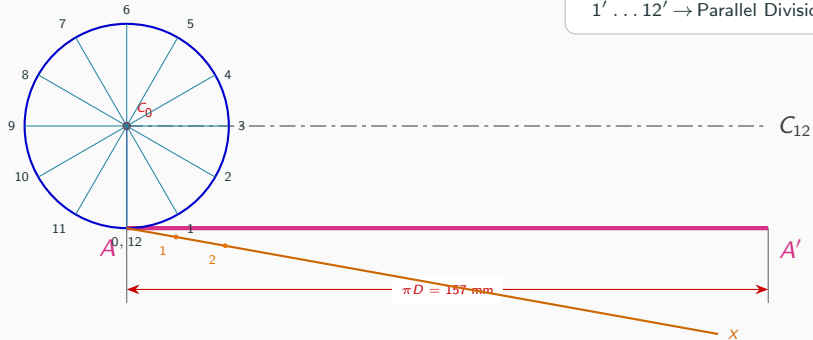
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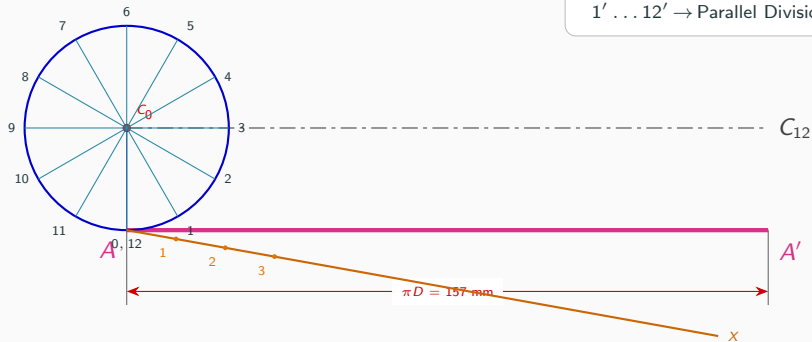
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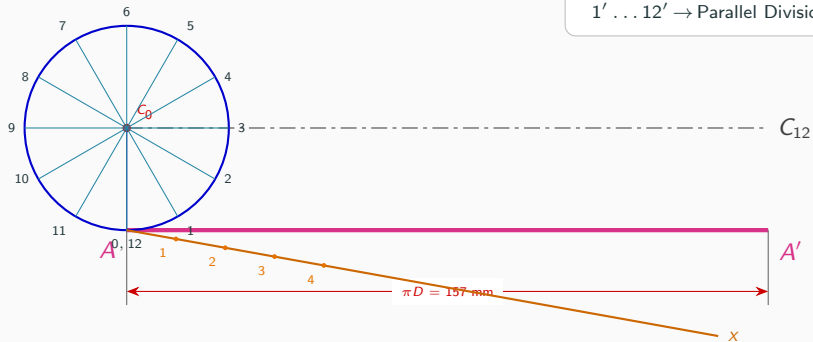
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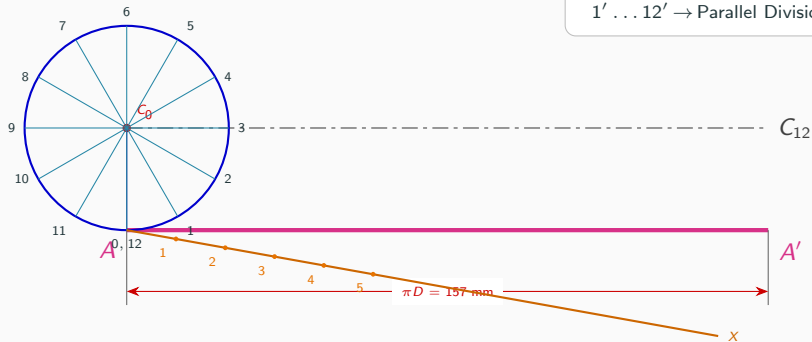
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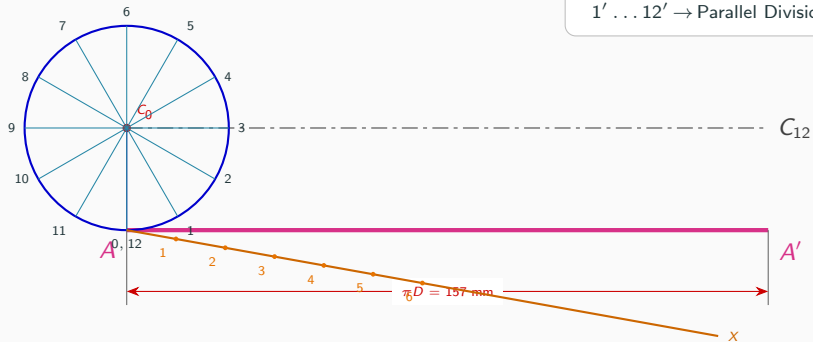
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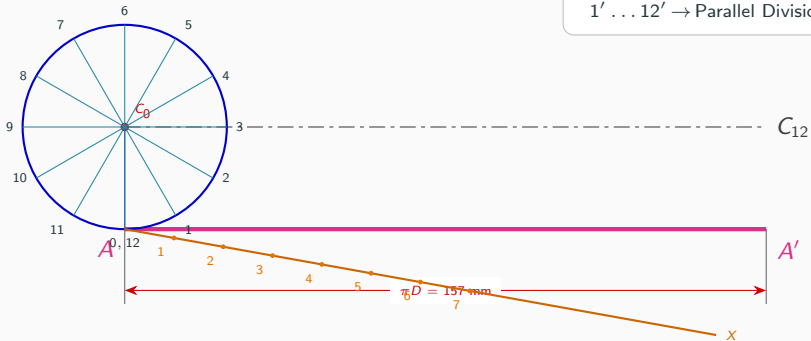
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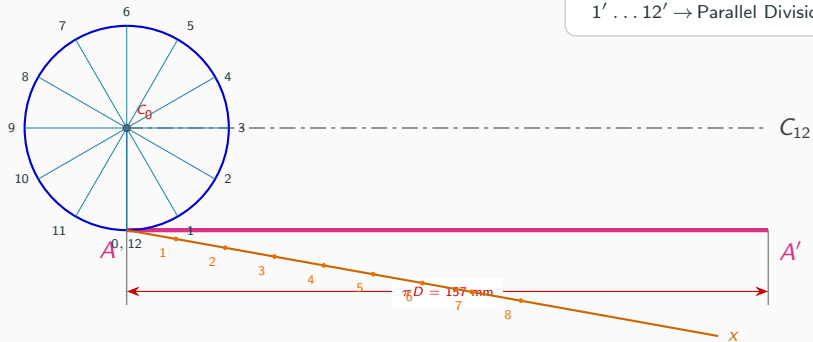
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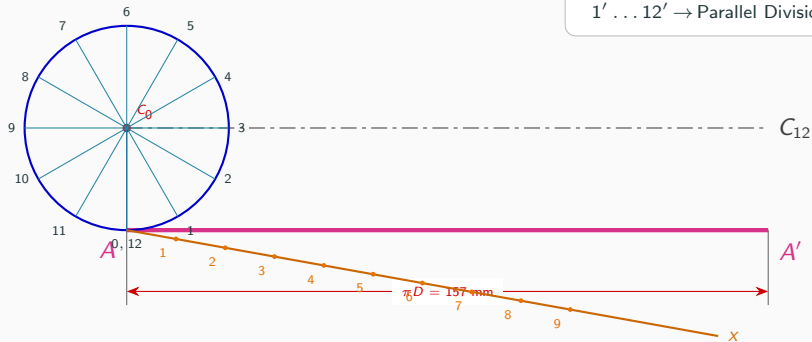
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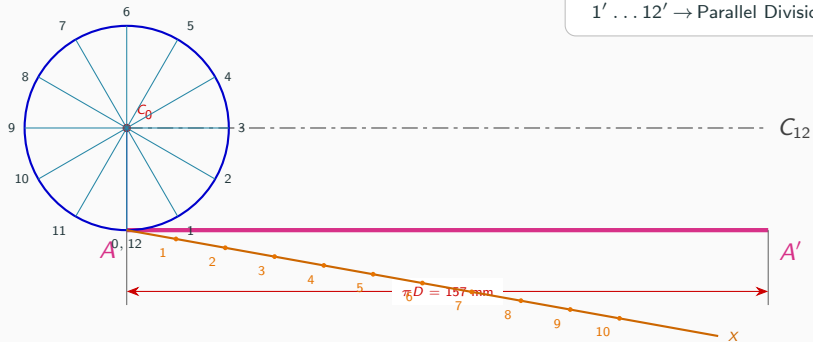
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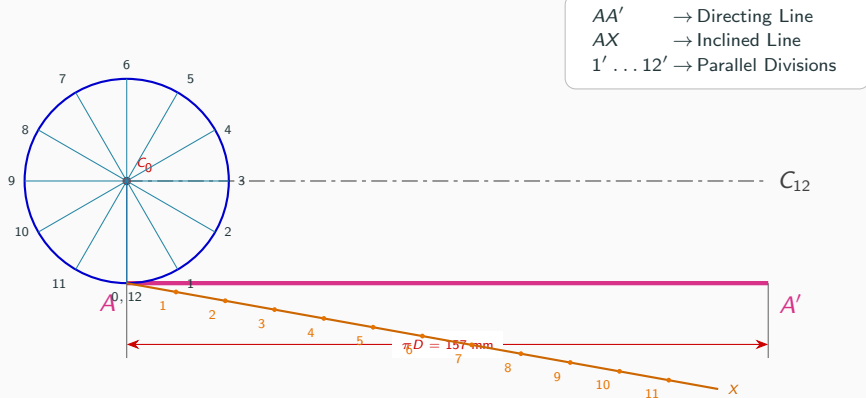
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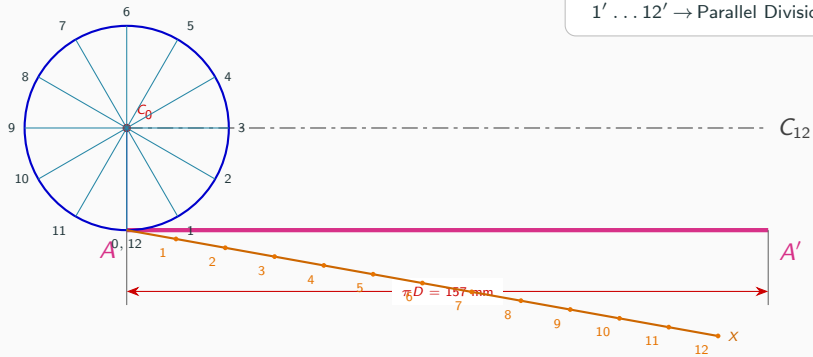
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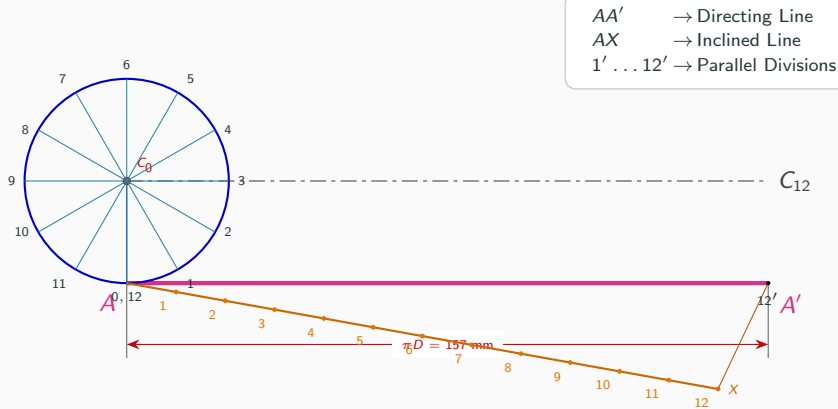
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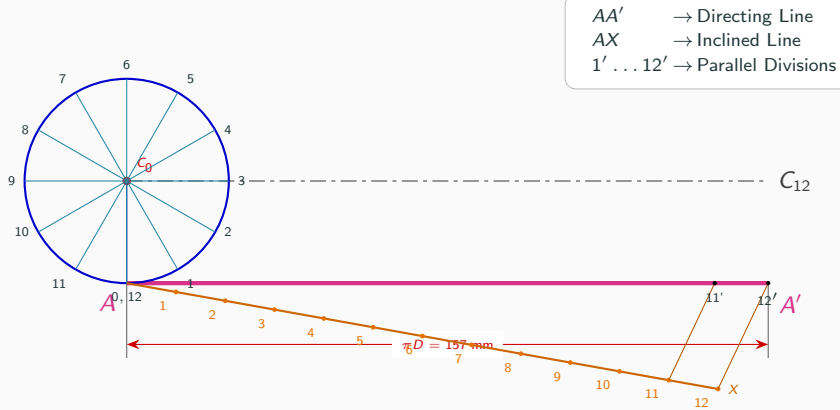
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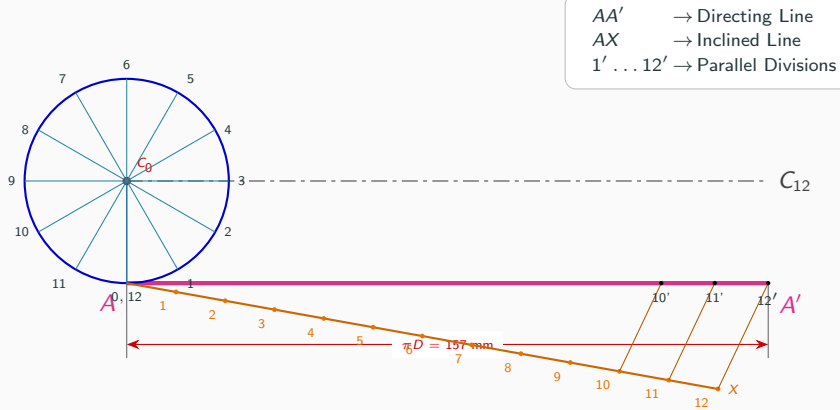
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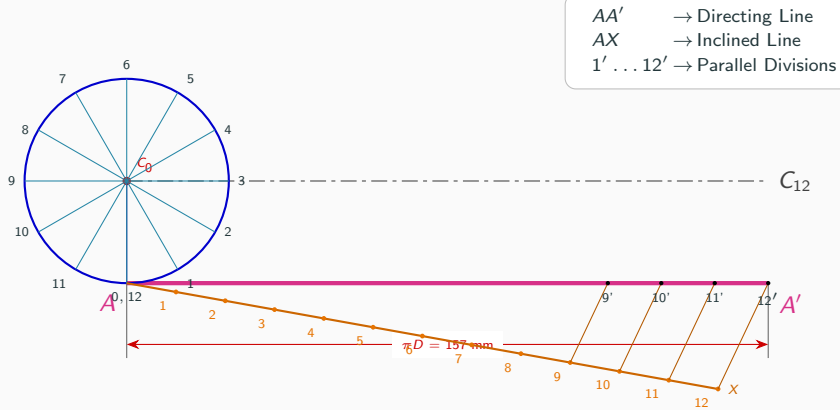
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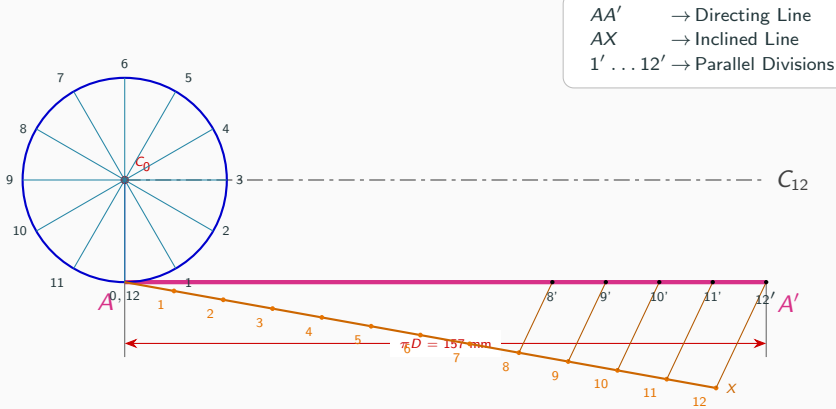
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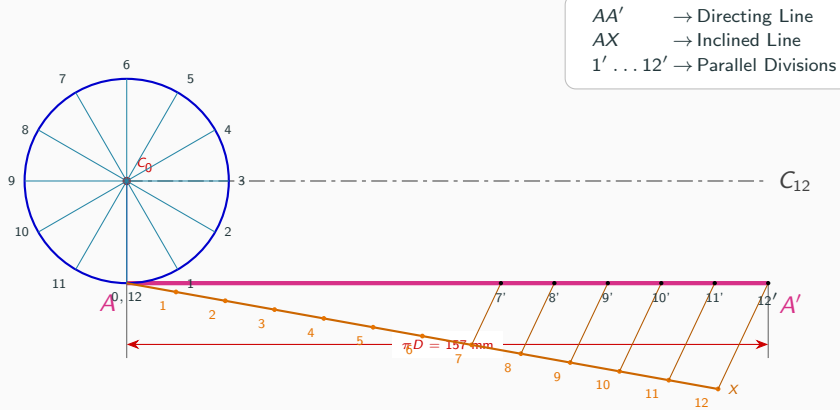
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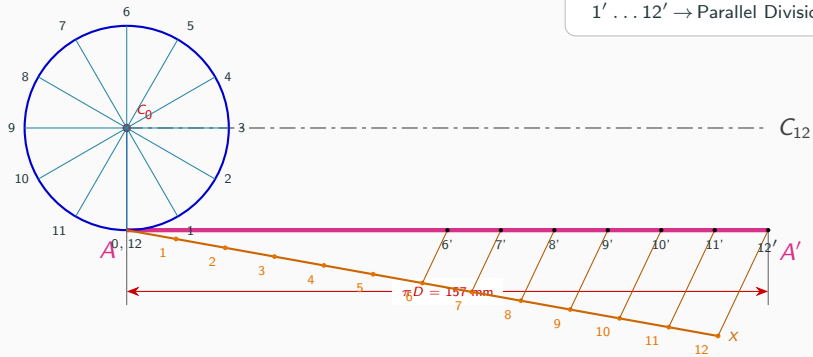
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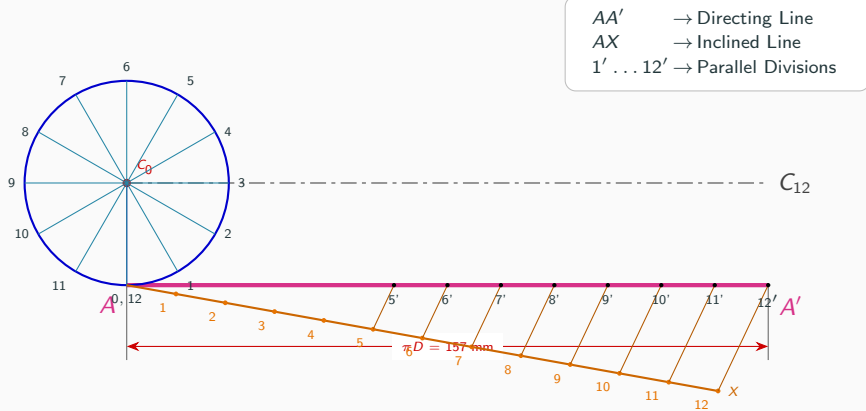
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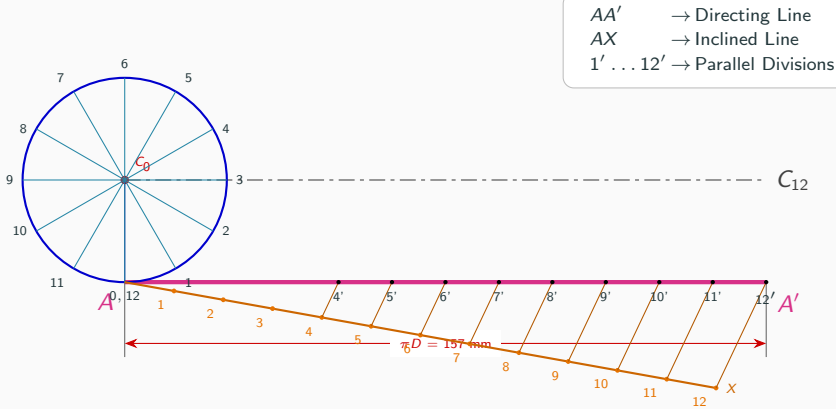
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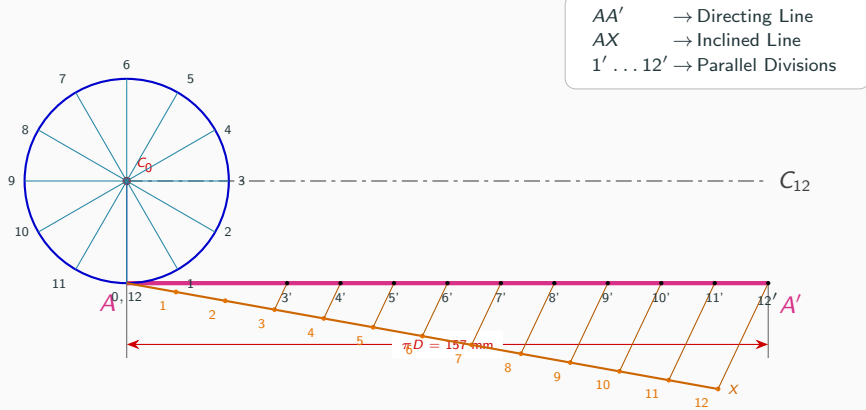
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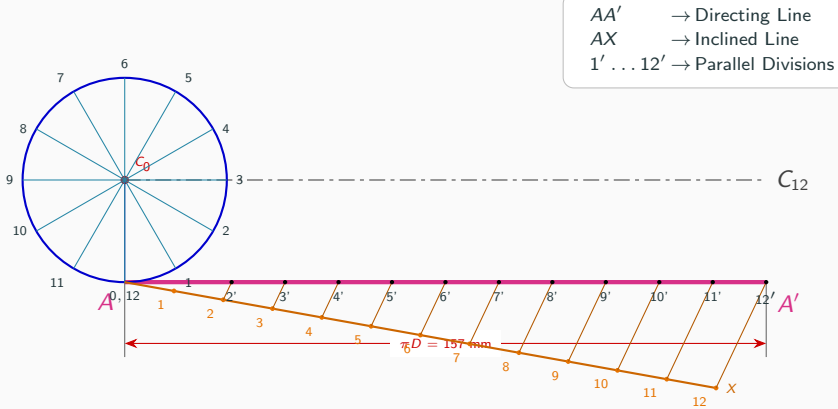
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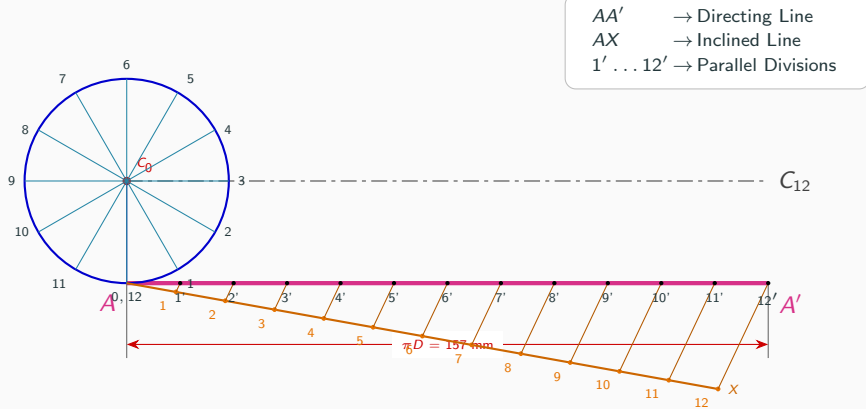
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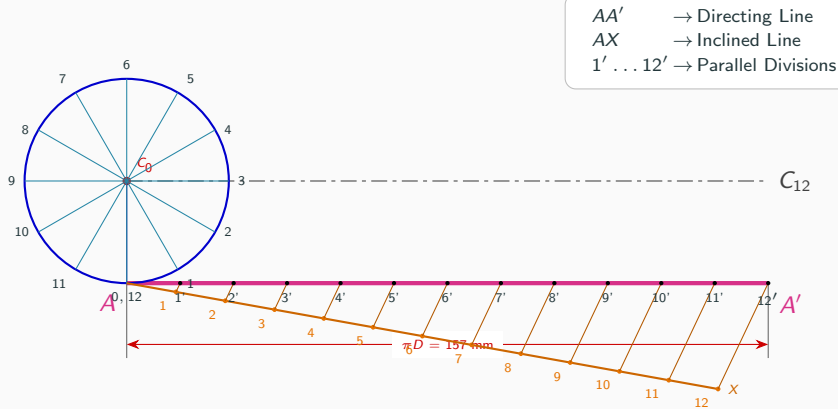
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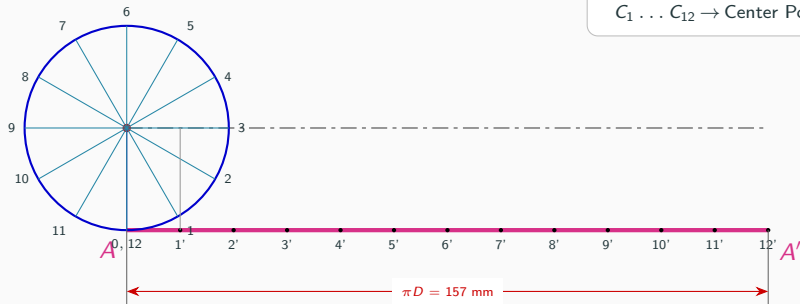
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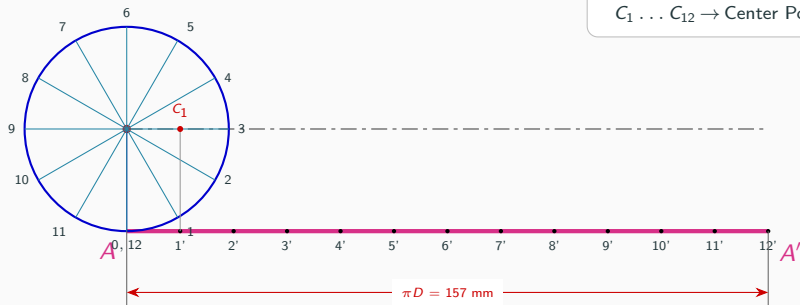
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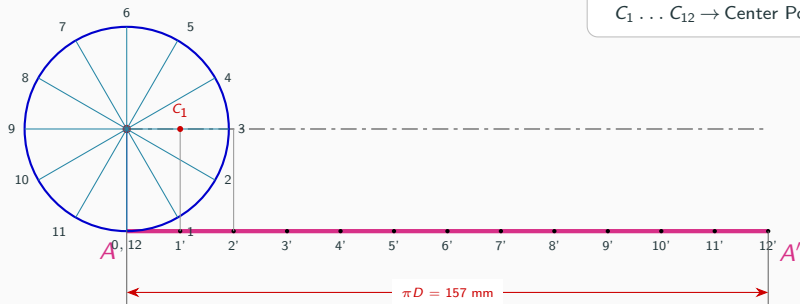
## Step 4: Project Divisions to Locate Centers $C_1 \dots C_{12}$



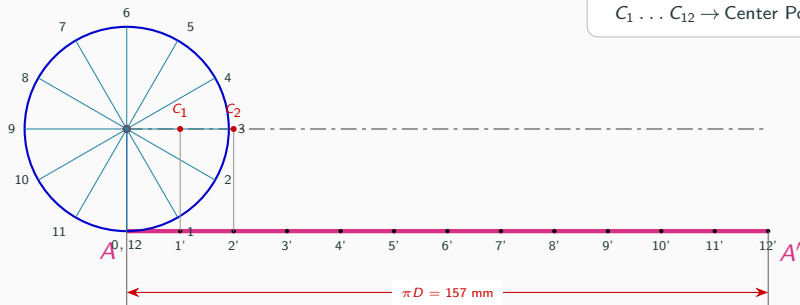
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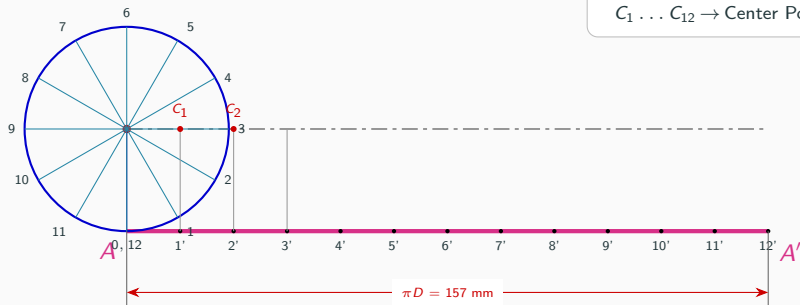
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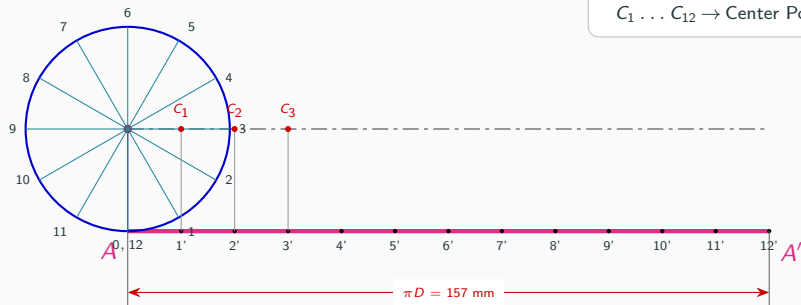
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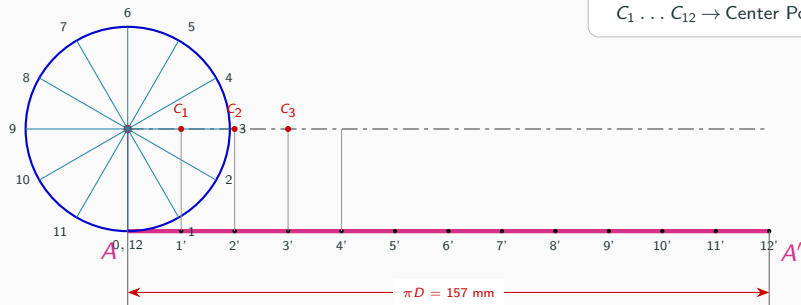
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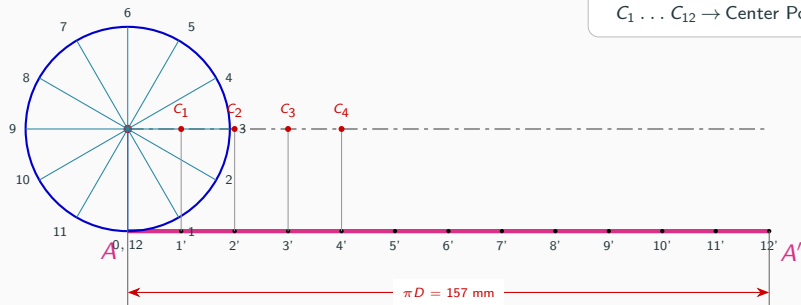
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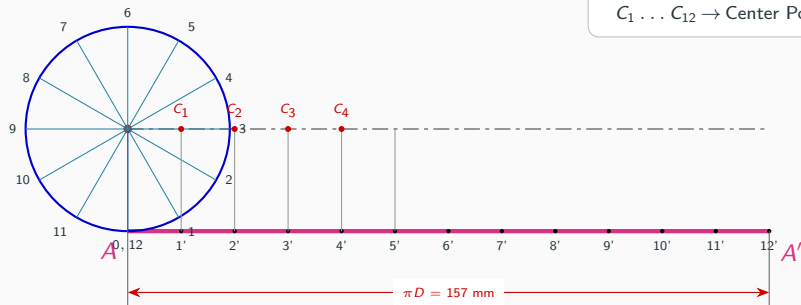
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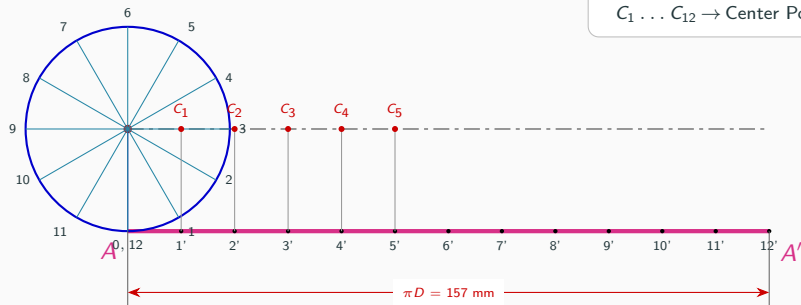
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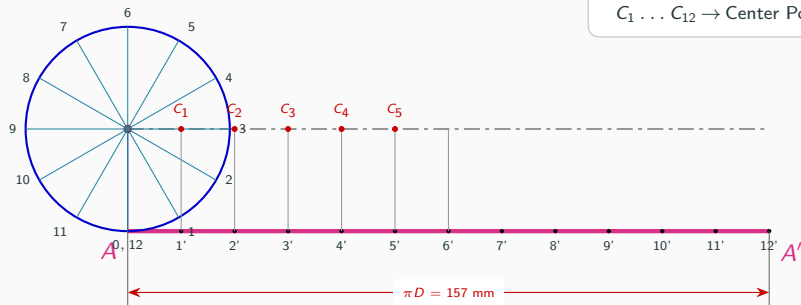
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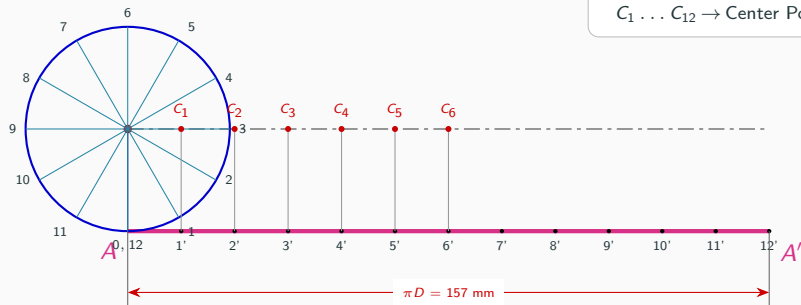
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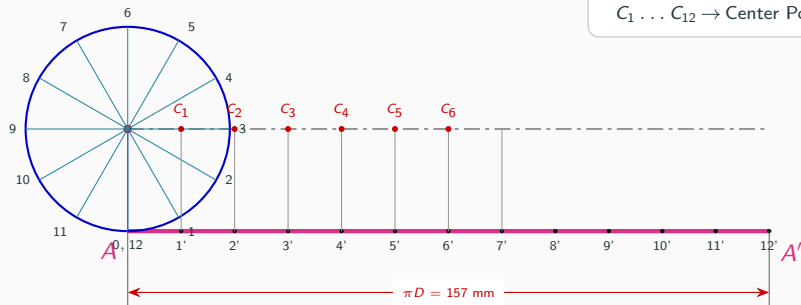
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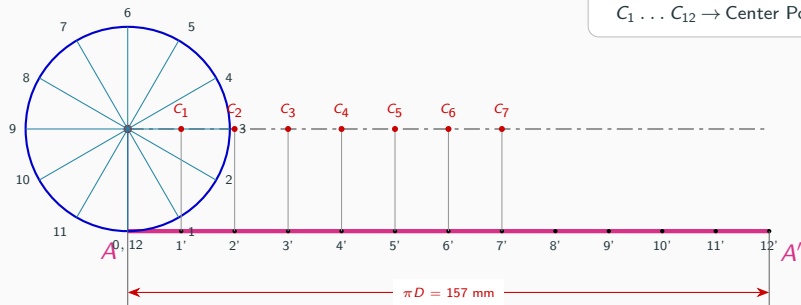
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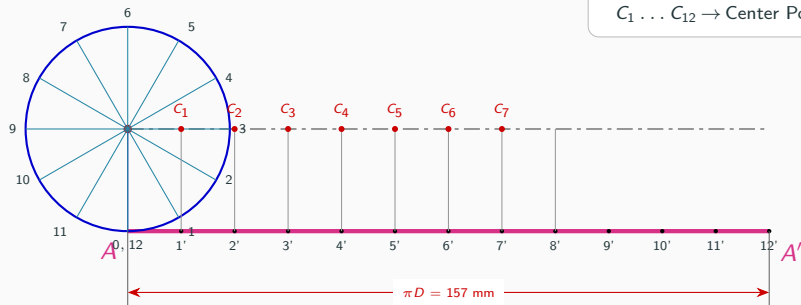
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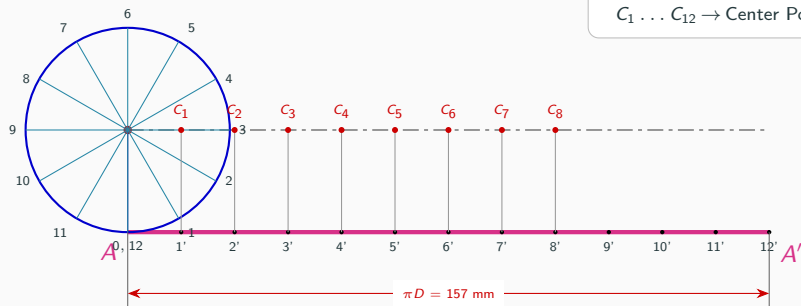
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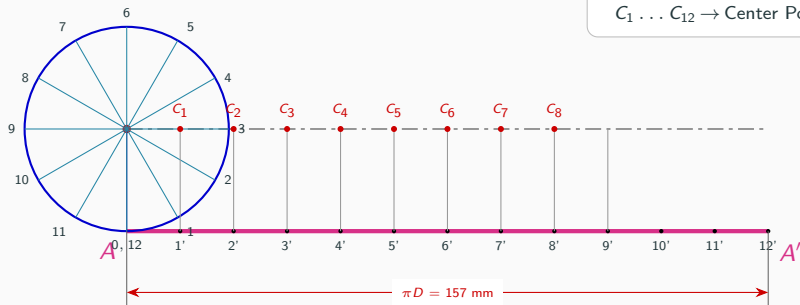
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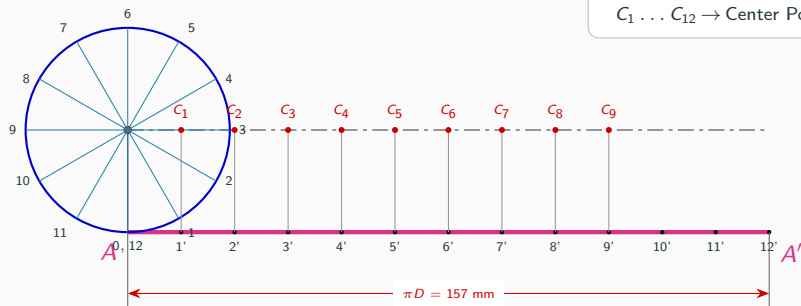
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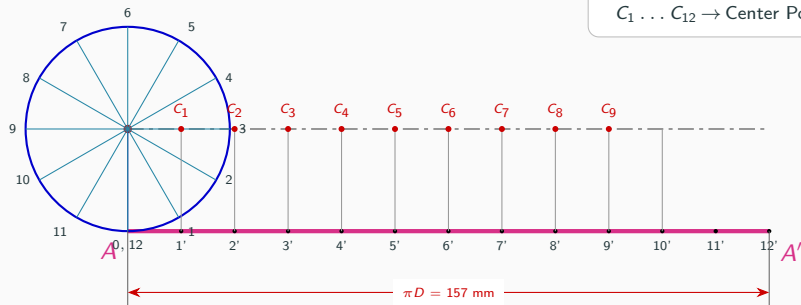
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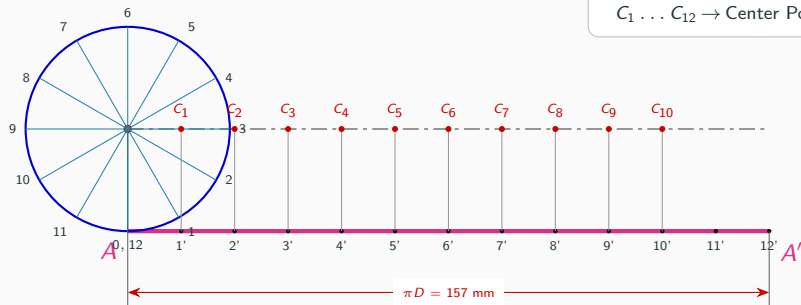
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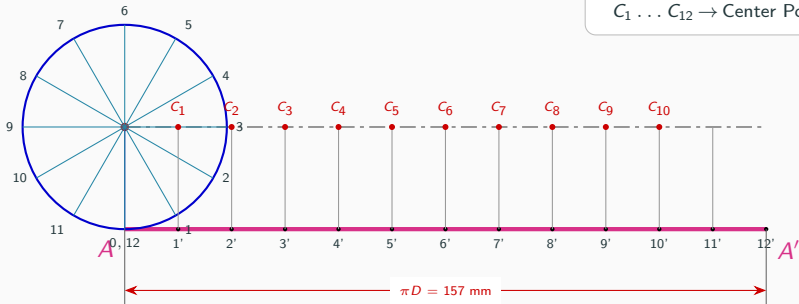
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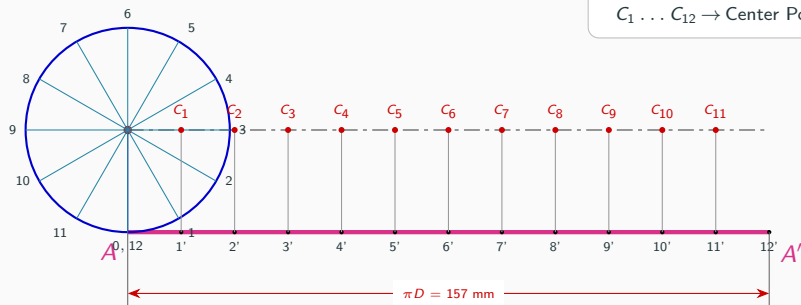
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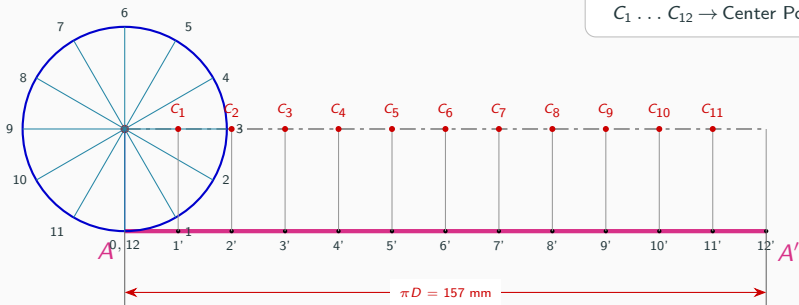
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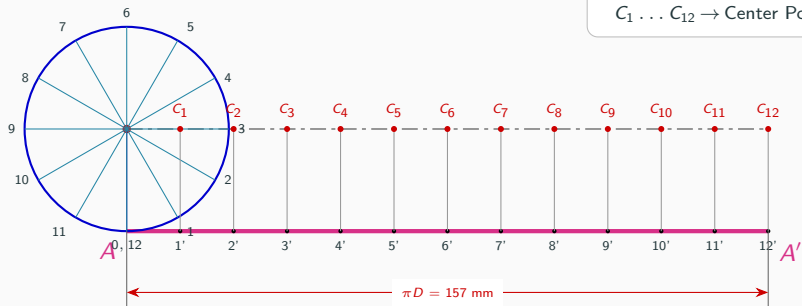


$AA'$  → Directing Line

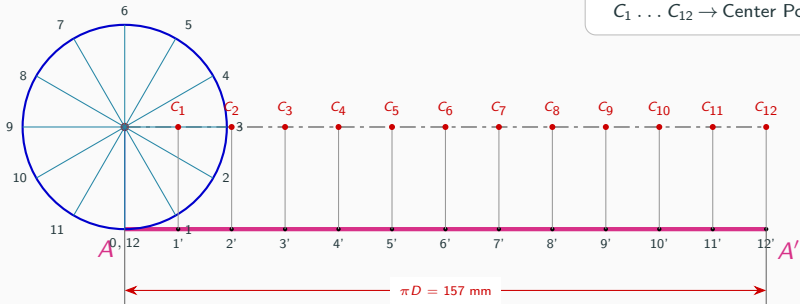
**Axis Line** → **Center Line**

$C_1 \dots C_{12} \rightarrow$  Center Points

## Step 4: Project Divisions to Locate Centers $C_1 \dots C_{12}$



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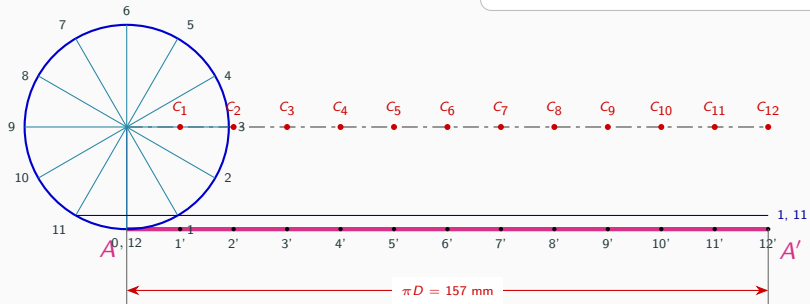


$AA'$  → Directing Line

**Axis Line** → **Center Line**

 $C_1 \dots C_{12} \rightarrow \text{Center Points}$

## Step 5: Draw Horizontal Reference Lines



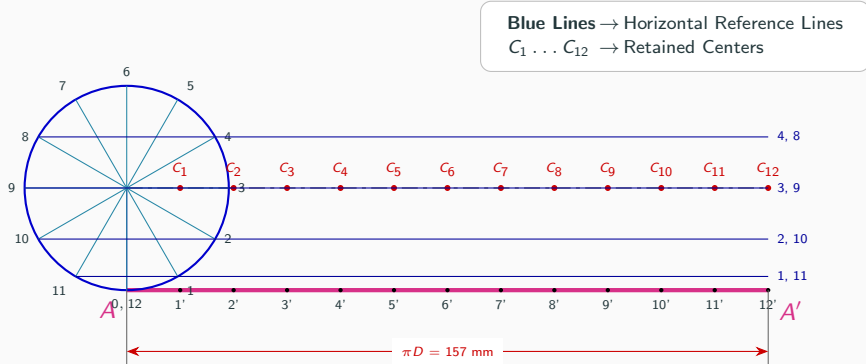
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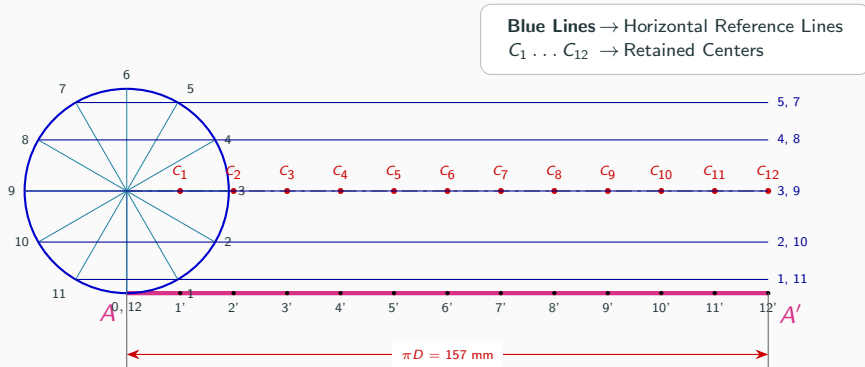
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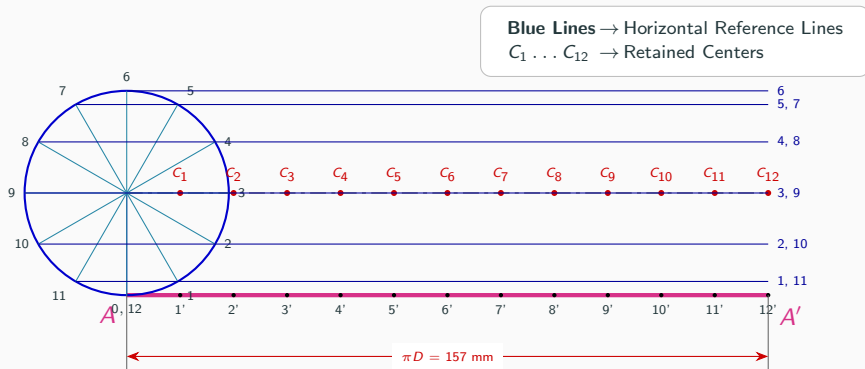
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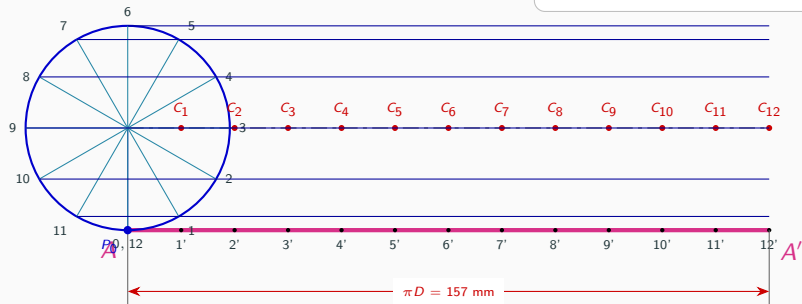
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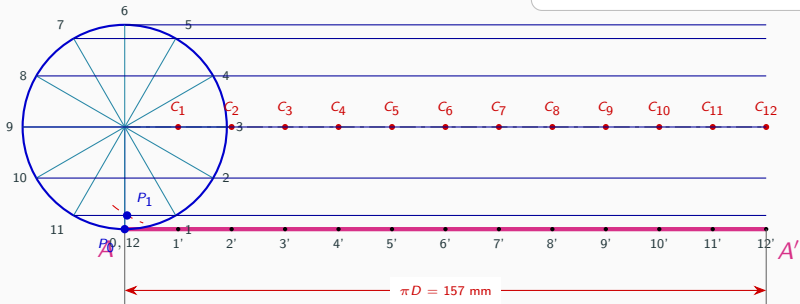
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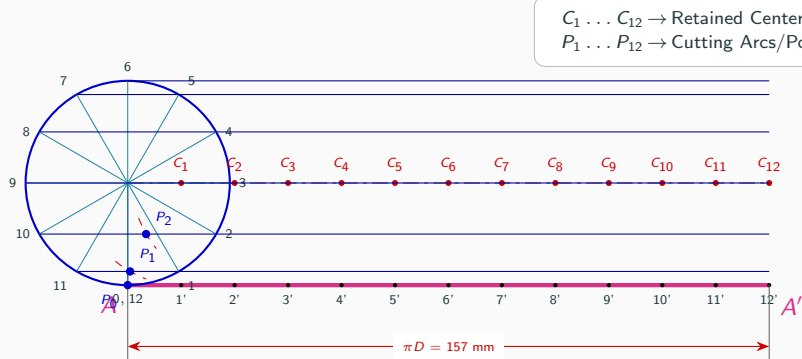
## Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$



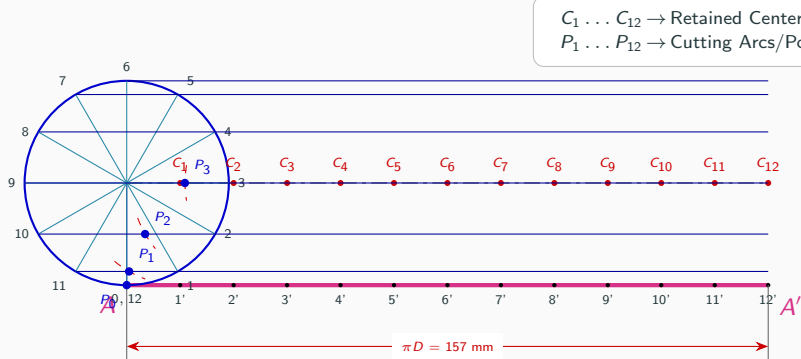
## Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$



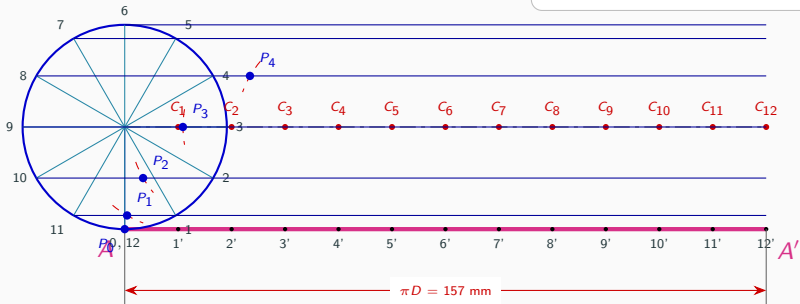
## Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$



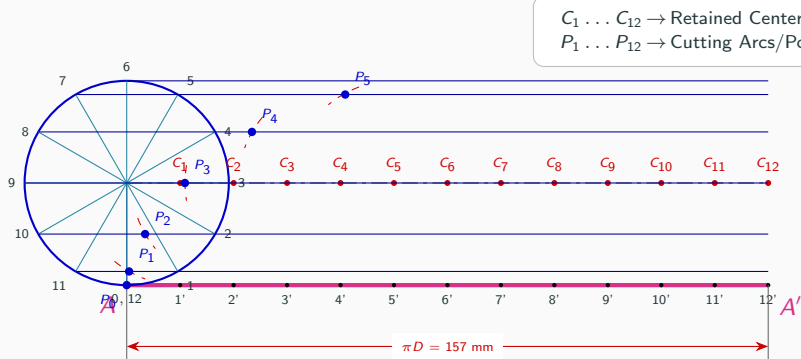
## Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$



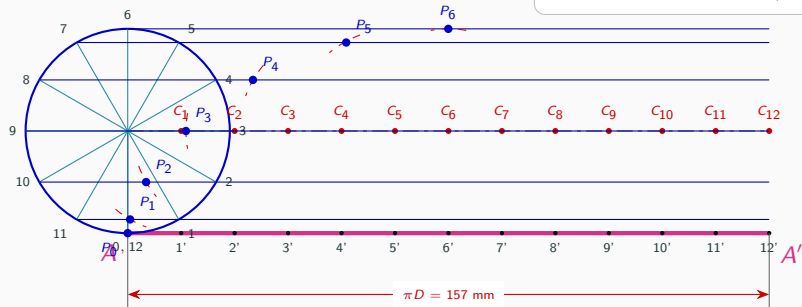
### Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$



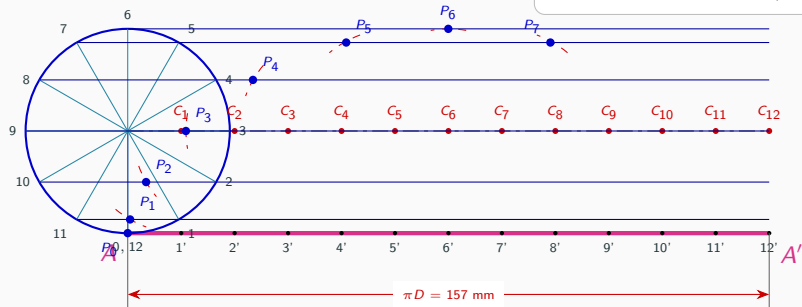
## Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$



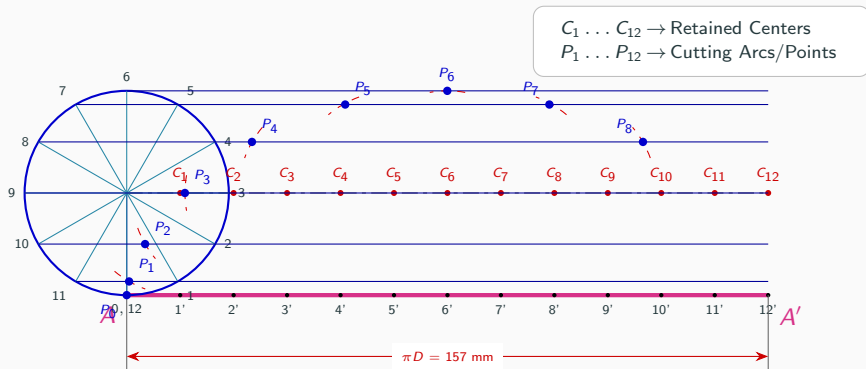
## Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$



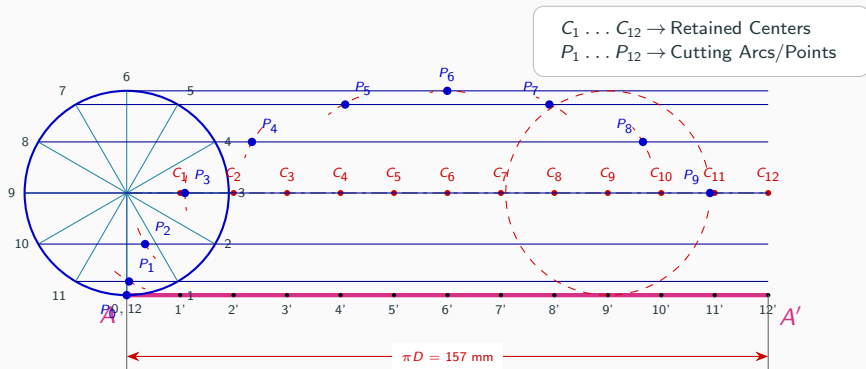
## Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$



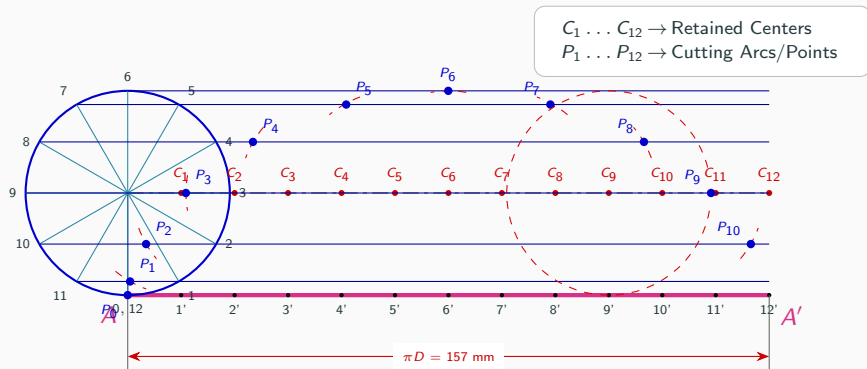
## Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$



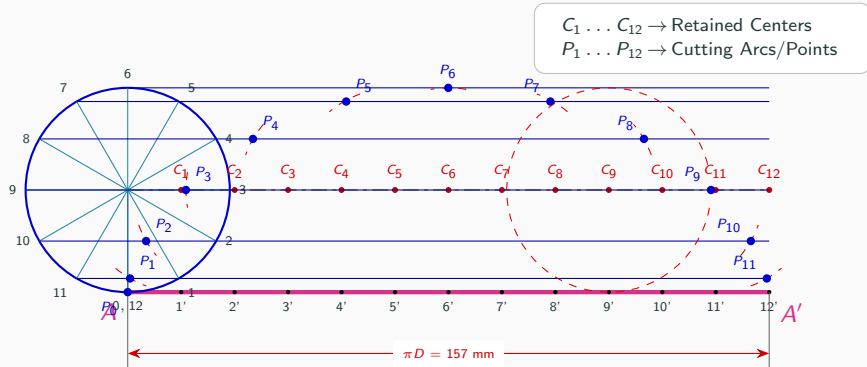
## Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$



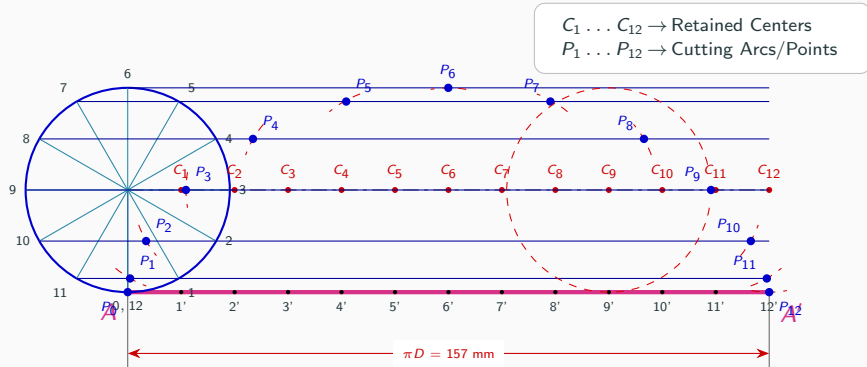
## Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$



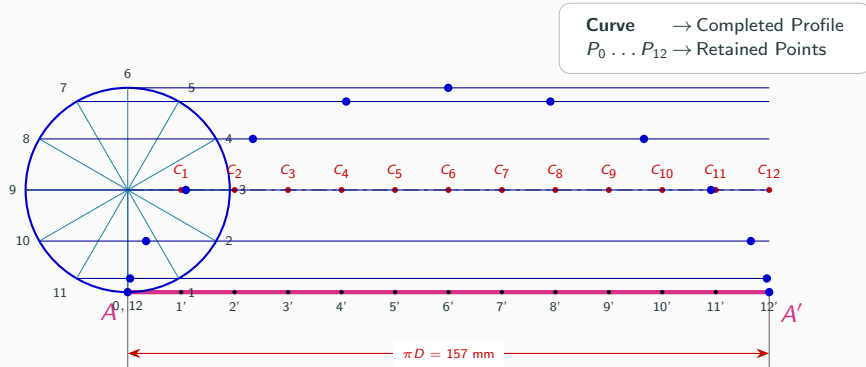
## Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$



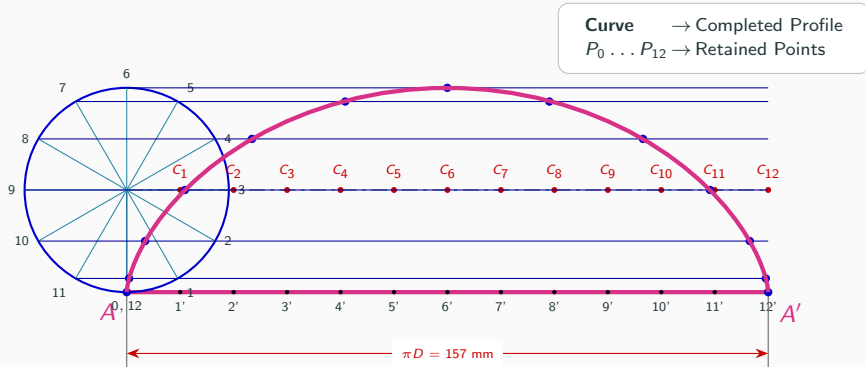
## Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$



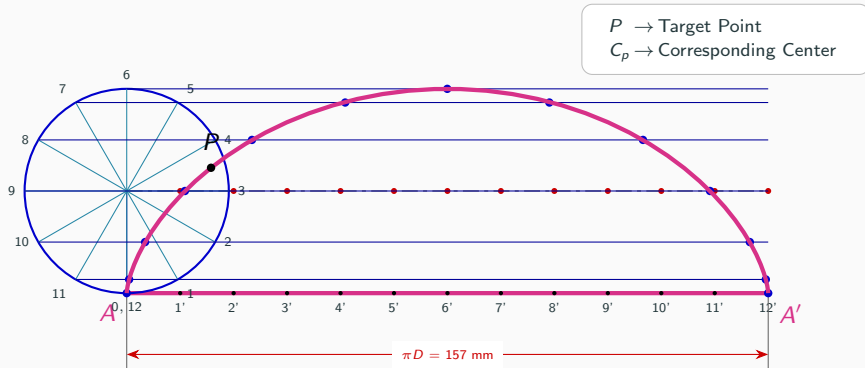
## Step 7: Smoothly Join Points to Form Cycloid Curve



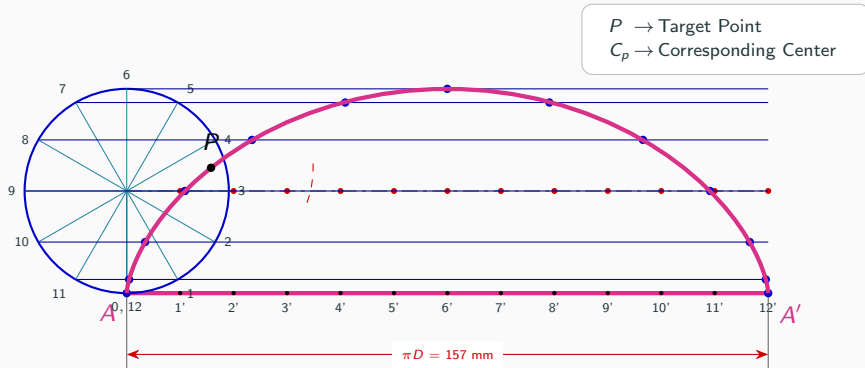
## Step 7: Smoothly Join Points to Form Cycloid Curve



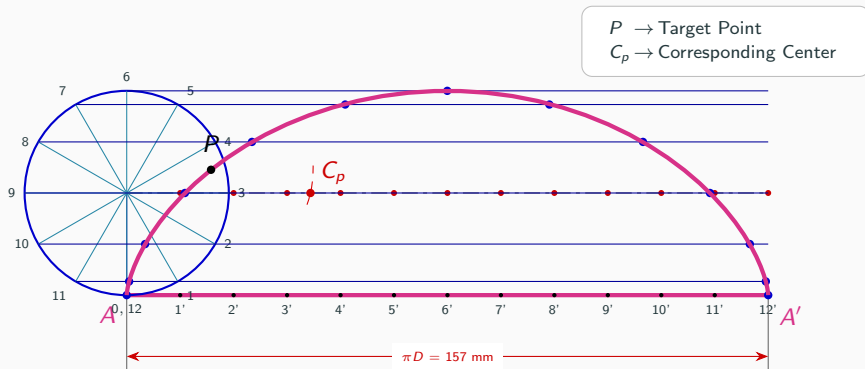
## Step 8: Mark Point $P$ & Locate Center $C_p$



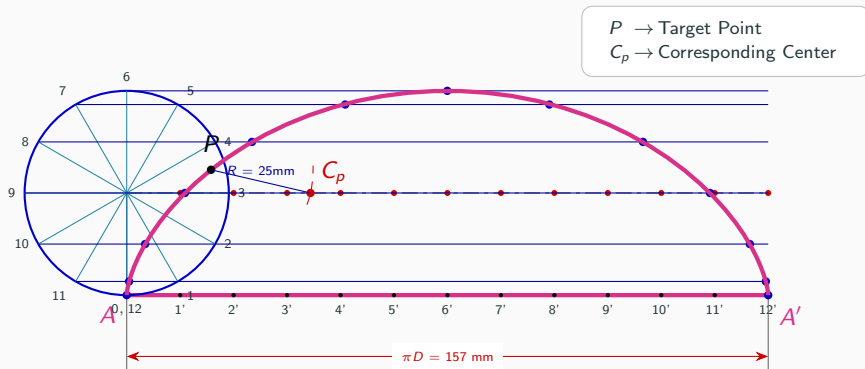
## Step 8: Mark Point $P$ & Locate Center $C_p$



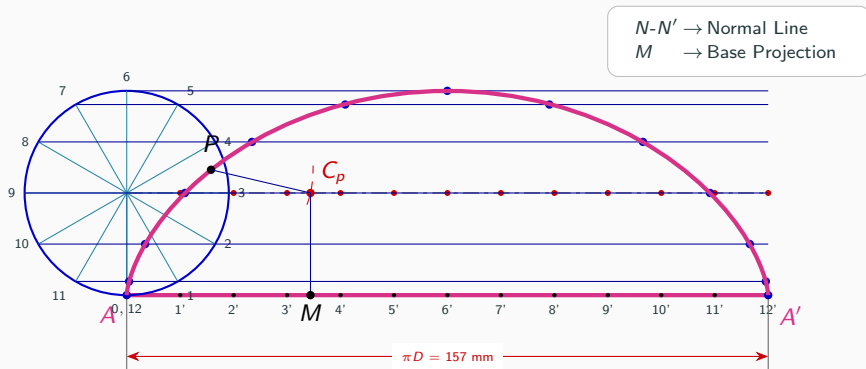
## Step 8: Mark Point $P$ & Locate Center $C_p$



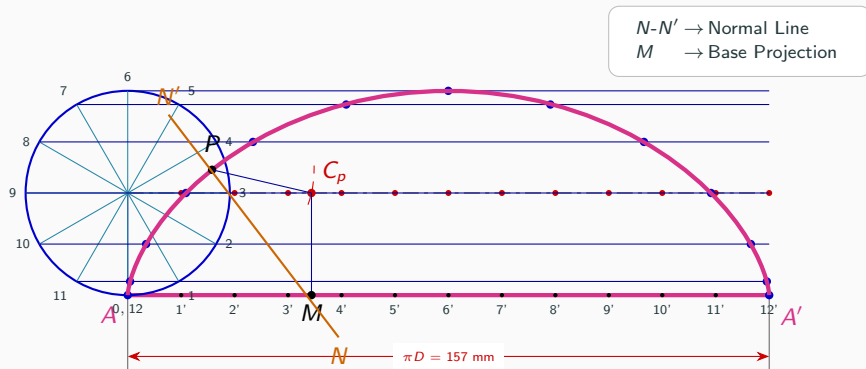
## Step 8: Mark Point $P$ & Locate Center $C_p$



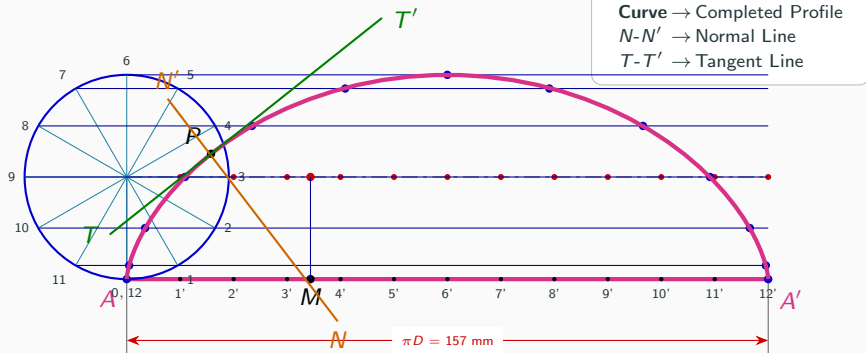
## Step 9: Construct Normal Line $N-N'$



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## Step 10: Construct Tangent Line $T-T'$ & Final Result



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